

CompliPilot

AI Compliance Copilot for Indian SMBs

System Design Document

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Apprend Technologies

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1. Executive Summary

CompliPilot is a unified AI-driven compliance platform designed to simplify regulatory obligations for Indian small and medium businesses and Chartered Accountant (CA) firms. Building on the foundation of GSTBot, CompliPilot extends coverage across the full spectrum of Indian regulatory requirements: RBI regulations, SEBI compliance, MCA filings, FEMA obligations, labor laws, and the Digital Personal Data Protection (DPDP) Act.

Indian SMBs face an increasingly complex regulatory environment with over 1,500 compliance obligations across central and state governments. Non-compliance penalties range from monetary fines to criminal prosecution. Most SMBs rely on manual tracking via spreadsheets and periodic CA consultations, leading to missed deadlines and costly penalties.

CompliPilot automates compliance tracking, filing generation, regulatory change monitoring, and deadline management through an AI-powered platform. The system ingests regulatory updates from government gazettes, circulars, and notifications, analyzes their impact on each client's specific obligations, and generates actionable alerts with recommended actions.

Target Users: Indian SMBs (direct compliance management), CA firms (managing compliance for multiple clients), and mid-size enterprises with dedicated compliance teams.

Key Differentiator: Unified multi-regulation coverage with AI-powered impact analysis, proactive regulatory change alerts, and automated filing generation, all built for the Indian regulatory landscape.

2. System Architecture

2.1 Technology Stack

Layer	Technology	Purpose
Runtime	Python 3.11+	Backend services, regulatory parsers
Backend Framework	FastAPI	REST API, background tasks
Frontend	React / Next.js 14	Compliance dashboard, filing editor
Primary Database	PostgreSQL 16	Compliance data, filing records, audit logs
Cache / Queue	Redis 7 + BullMQ	Deadline scheduler, notification queue
Vector Store	Qdrant	Regulatory text embeddings for semantic search
AI Gateway	OpenRouter	LLM for document analysis and filing generation
Document Parser	Apache Tika + Custom	PDF/gazette parsing pipeline
Notification	WhatsApp + Email + SMS	Multi-channel compliance alerts

2.2 Regulatory Data Pipeline

The regulatory data pipeline continuously monitors government sources for new circulars, notifications, and amendments:

1. **Gazette Scraper:** Automated scraping of official sources including the Official Gazette of India, MCA portal, RBI circulars, SEBI notifications, and state government bulletins.
2. **Document Parser:** Extracts structured data from PDFs, HTML pages, and scanned documents using OCR where necessary.
3. **Regulatory Analyzer:** LLM-powered analysis that classifies the regulatory update by domain, identifies affected entity types, and extracts key compliance requirements.
4. **Vector Indexing:** Regulatory text is chunked and embedded into Qdrant for semantic search and contextual retrieval during impact analysis.
5. **Impact Mapper:** Cross-references new regulations against each client's profile (entity type, industry, state, turnover) to generate personalized impact assessments.
6. **Alert Dispatcher:** Sends targeted notifications via WhatsApp, email, and SMS based on client notification preferences and urgency level.

2.3 Multi-tenant Architecture

CompliPilot supports a CA firm to client hierarchy. A CA firm is the primary tenant with multiple client organizations underneath. Each client organization has its own set of compliance obligations, filings, and documents. CA firm users can switch between client contexts seamlessly.

PostgreSQL Row-Level Security (RLS) enforces tenant isolation at the database level. Every query includes `organization_id` filtering. Redis keys follow the pattern `complipilot:{orgId}:{resource}:{id}`.

3. Data Model

3.1 Core Tables

Table	Description	Key Fields
organizations	CA firms and client companies	id, name, type (ca_firm/company), gstin, pan, cin, state, industry
clients	CA firm to client mapping	id, ca_firm_id, org_id, engagement_type, start_date, status
compliance_obligations	Regulatory requirements catalog	id, regulation, section, description, frequency, entity_types_json
filings	Individual filing instances	id, org_id, obligation_id, due_date, status, submitted_at, acknowledgement_no
deadlines	Calculated deadline calendar	id, org_id, filing_id, due_date, reminder_sent, escalation_level
regulatory_updates	Parsed regulatory changes	id, source, date, title, summary, impact_level, domains_json
documents	Uploaded and generated documents	id, org_id, type, storage_path, parsed_content, metadata_json
audit_trails	Tamper-proof activity log	id, org_id, user_id, action, entity_type, entity_id, timestamp, checksum
notifications	Sent notification records	id, org_id, channel, recipient, content, sent_at, status
templates	Filing and document templates	id, regulation, filing_type, template_json, version, is_active

The audit_trails table is append-only with cryptographic checksums for tamper detection. Each entry includes a SHA-256 hash of the previous entry, creating an audit chain. No UPDATE or DELETE operations are permitted on this table.

4. Core Features

4.1 Regulatory Calendar with Auto-Reminders

A comprehensive compliance calendar showing all upcoming deadlines across all regulations. Deadlines are color-coded by urgency (green > 30 days, yellow 15-30 days, orange 7-15 days, red < 7 days). Automatic reminders are sent via WhatsApp and email at configurable intervals (default: 30, 15, 7, 3, 1 days before deadline).

4.2 AI-Generated Compliance Filings

CompliPilot generates draft filings by pulling data from the organization's profile, previous filings, and financial records. Supported filing types include: MCA Annual Returns (MGT-7, AOC-4), GST returns (GSTR-1, GSTR-3B, GSTR-9), ROC filings (INC-20A, DIR-3 KYC, ADT-1), and Income Tax returns (ITR-4, ITR-6). Generated drafts are presented for human review before submission.

4.3 Document Parser for Regulatory Notices

Upload regulatory notices, show-cause notices, or circulars and CompliPilot will parse the document, extract key requirements, identify deadlines, and suggest response actions. The parser handles PDFs (including scanned), Word documents, and HTML pages.

4.4 DPDP Act Compliance Toolkit

A dedicated module for Digital Personal Data Protection Act compliance:

1. Consent Management: Template-based consent collection with versioning and audit trail.
2. Data Mapping: Automated discovery and mapping of personal data across the organization's systems.
3. Breach Notification: Workflow for data breach detection, assessment, DPA notification (within 72 hours), and affected individual notification.
4. Data Subject Rights: Automated handling of access, correction, and erasure requests with SLA tracking.
5. Privacy Impact Assessment: Guided PIA templates with AI-assisted risk scoring.

4.5 Multi-Entity Management for CA Firms

CA firms can manage compliance for hundreds of client entities from a single dashboard. Features include bulk deadline tracking, cross-client filing status overview, workload distribution among CA firm staff, and consolidated billing. Client-level access controls ensure staff can only see assigned clients.

4.6 Regulatory Change Alerts with Impact Analysis

When a new regulation or amendment is detected, CompliPilot performs an automated impact analysis for each client organization. The alert includes: what changed, who is affected, what action is required, and the deadline for compliance. Alerts are prioritized by impact level (critical, high, medium, low) and sent via the client's preferred notification channel.

4.7 Audit Trail with Tamper-Proof Logging

Every action in the system is logged with a cryptographic hash chain. The audit trail captures who did what, when, from which IP, and the before/after state of modified records. This creates an immutable compliance history that can be presented to regulators during audits.

4.8 Template Library

A curated library of templates for common filings, board resolutions, compliance certificates, and regulatory responses. Templates are maintained by the CompliPilot team and updated when regulations change. CA firms can create custom templates for their practice.

4.9 GSTBot Integration

Native integration with GSTBot for GST-specific compliance. Invoice reconciliation, ITC matching, e-way bill management, and return filing are handled by GSTBot's specialized engine, with status and deadlines surfaced in CompliPilot's unified dashboard.

5. Regulatory Coverage Matrix

Regulation	Filing Types	Frequency	Key Deadlines
GST	GSTR-1, GSTR-3B, GSTR-9, GSTR-9C	Monthly/Annual	11th, 20th of each month; Dec 31 for annual
Income Tax	ITR-1 to ITR-7, TDS Returns, Advance Tax	Quarterly/Annual	Jul 31, Oct 31, Mar 15 (advance tax)
RBI	FLA Return, ECB Returns, Annual Return on Foreign Liabilities	Annual/Event-based	Jul 15 (FLA), event-based for ECB
SEBI	Shareholding Pattern, Board Meeting Intimation, Insider Trading	Quarterly/Event	21 days post quarter-end
MCA/ROC	MGT-7, AOC-4, DIR-3 KYC, ADT-1, INC-20A	Annual/Event	Oct 31 (MGT-7), Nov 30 (AOC-4)
FEMA	FC-GPR, FC-TRS, Annual Return on FDI	Event-based/Annual	30 days from event; Jul 15 annual
Labor Laws	PF Returns, ESI Returns, Professional Tax, Shops & Est.	Monthly/Annual	15th of each month (PF/ESI)
DPDP Act	Consent Records, Breach Notification, DPA Annual Report	Ongoing/Event	72 hours (breach); annual report

The platform covers over 200 specific compliance obligations across these eight regulatory domains. Coverage is expandable through the template system and regulatory update pipeline.

6. API Design

6.1 RESTful Endpoints

Method	Endpoint	Description
GET	/api/v1/obligations	List compliance obligations for an organization
GET	/api/v1/obligations/:id/deadlines	Get upcoming deadlines for an obligation
POST	/api/v1/filings	Create a new filing draft
PUT	/api/v1/filings/:id	Update filing data
POST	/api/v1/filings/:id/submit	Submit filing (after human review)
GET	/api/v1/calendar	Compliance calendar with filters
POST	/api/v1/documents/parse	Upload and parse a regulatory document
GET	/api/v1/alerts	List regulatory change alerts
GET	/api/v1/alerts/:id/impact	Get impact analysis for an alert

GET	/api/v1/templates	List available filing templates
GET	/api/v1/audit-trail	Query audit trail with filters
POST	/api/v1/dpdp/breach-report	Initiate breach notification workflow

All endpoints require JWT authentication with organization context. CA firm tokens include a client_org_id claim for cross-organization access. Rate limiting is enforced per organization.

7. Revenue Model

7.1 Pricing Tiers

Plan	Monthly Fee	Entities	Regulations	Features
SMB	INR 2,999/mo	1	GST + Income Tax + MCA	Calendar, reminders, basic filings
Professional	INR 9,999/mo	Up to 25	All regulations	Multi-entity, filing generation, DPDP toolkit
CA Firm	INR 19,999/mo	Up to 100	All regulations	Staff management, bulk operations, client portal
Enterprise	INR 29,999/mo	Unlimited	All + custom	API access, custom integrations, dedicated support

Annual billing receives a 20% discount. Additional entities can be added at INR 299/entity/month on the Professional plan and INR 199/entity/month on the CA Firm plan. All plans include WhatsApp and email notifications.

8. Security

8.1 Data Protection

All data is encrypted at rest using AES-256 and in transit using TLS 1.3. Database backups are encrypted and stored in a geographically separate location within India (data residency requirement). PII fields (PAN, Aadhaar, GST numbers) are encrypted at the column level using application-layer encryption.

8.2 Access Control

Role-based access control (RBAC) with four default roles: Admin, Compliance Manager, Staff, and Read-Only. CA firms can define custom roles. All role changes are logged in the audit trail. Two-factor authentication (TOTP) is mandatory for Admin and Compliance Manager roles.

8.3 Audit and Compliance

The platform itself is designed to be audit-ready. The tamper-proof audit trail, encrypted storage, access controls, and data residency compliance ensure that CompliPilot meets the requirements it helps clients fulfill. SOC 2 Type II certification is planned for Year 2.

9. Testing Strategy

9.1 Test Categories

Category	Scope	Tools	Coverage Target
Unit Tests	Regulatory rule engine,	pytest, unittest.mock	>90% line coverage

	deadline calculator		
Integration Tests	Filing generation, document parsing	pytest + testcontainers	>80% path coverage
Regulatory Tests	Deadline accuracy, filing format validation	Custom rule test suite	100% of supported filings
Security Tests	Auth, RBAC, encryption, audit integrity	OWASP ZAP, custom scripts	All OWASP Top 10
E2E Tests	Full filing workflow from draft to submission	Playwright	All critical workflows

9.2 Regulatory Rule Testing

Each regulatory rule (deadline calculation, filing requirement, entity applicability) has dedicated test cases derived from the actual regulation text. Tests are updated whenever a regulatory change affects the rule. A regulatory test failure blocks deployment.

10. Milestones

Phase	Timeline	Deliverables	Success Criteria
MVP	Weeks 1-6	GST + MCA compliance, calendar, reminders, basic filings	20 SMB pilot users, zero missed deadlines
Phase 2	Weeks 7-10	RBI + SEBI compliance, CA firm multi-entity, regulatory alerts	5 CA firms onboarded, 50+ entities managed
Phase 3	Weeks 11-12	DPDP toolkit, labor law compliance, template library	Full regulatory coverage operational
Beta	Weeks 13-14	Filing generation, document parser, GSTBot integration	100+ active organizations
GA	Weeks 15-16	Production hardening, billing, onboarding automation	Public launch with SLA

The MVP focuses on GST and MCA compliance as these represent the highest-volume obligations for Indian SMBs. Each subsequent phase adds regulatory domains based on client demand and market research.

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